

Lab 2: Basic Searching with Splunk Search Language

Learning Objectives

After completing this lab, you will be able to:

- Perform basic searches using Splunk Search Language
- Use Boolean operators (AND, OR) and wildcards in searches
- Navigate and analyze search results effectively
- Use the timeline to identify patterns and trends
- Refine searches using field values
- Save and share search results with proper permissions
- Manage search jobs and histories

Prerequisites

- Splunk environment running and accessible
- Sample data loaded (completed Lab 1)
- Basic understanding of Splunk interface

Lab Scenario

There is reason to believe there might be a security issue with your web server. Your manager has asked you to explore failed SSH login attempts and identify potential security threats.

Data Sources

This lab uses the following data types:

- **Web Application** (sourcetype: `access_combined_wcookie`)
- **Database** (sourcetype: `db_audit`)
- **Web Server** (sourcetype: `linux_secure`)

Data window: The course dataset covers roughly **26 days** ending at the moment

`labs/data/generate_course_data.py` was last run — deliberately sized to sit inside Splunk's **Last 30 days** preset. Set the time picker to **Last 30 days** for every exercise unless a step says otherwise.

Regenerate the data within **4 days** of delivering the class; past that the preset starts clipping the oldest events and the answer-key counts drift.

Task 1: Perform a Basic Search

Steps:

1. Navigate to Search Interface

- If in Home app, click **Search & Reporting** from the left sidebar
- Alternatively, click **Search** from the top navigation bar

2. Execute Your First Search

- In the search bar, type: `error OR fail*`
- Set time range to **Last 30 days** using the time picker
- Click the **Search** button

3. Analyze Results

- Review search results and observe highlighted search terms
- Use pagination to browse through results
- Note the host, source, and sourcetype values at the bottom of each event
- Observe that most results come from `web_server` host (Linux secure logs)
- Notice the search terms are highlighted in yellow within the events

Validation:

- ☐ Search executed successfully
 - ☐ Results show highlighted search terms ("error", "fail", "failed", "failure")
 - ☐ Events primarily from `web_server` host with `linux_secure` sourcetype
 - ☐ Can navigate through paginated results
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Task 2: Narrow Your Search Results

Steps:

1. Start New Search

- Click **Search** to start fresh
- Enter search: `fail* AND password`
- Set time range to **Last 30 days**
- Execute search

2. Add Port Filter

- Note port values in several events
- At end of search string, add: 22
- Execute search
- Observe that events with any occurrence of "22" are returned

3. Refine with Exact Phrase

- Replace 22 with "port 22" (include quotes)
- Execute search
- Verify only events with exact phrase "port 22" appear
- Page through results to see login failures

Validation:

- ☐ Initial search returns password-related failures
 - ☐ Adding "22" returns too many results
 - ☐ Using "port 22" in quotes returns specific SSH login attempts
 - ☐ Multiple login failures visible
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Task 3: Analyze Trends Using Timeline

Steps:

1. Examine Timeline

- Look for patterns and spikes in the timeline above results
- Single-click on different columns to view events for specific time periods

2. Identify Attack Patterns

- Look for spikes that might indicate coordinated attacks
- Single-click another column and compare event patterns
- Note any similarities in IP addresses or user accounts

3. Time Range Drilling

- If you see a spike, double-click that column to zoom into that time range
- Analyze the concentrated activity during that period

Validation:

- ☐ Can interact with timeline columns
- ☐ Identified time periods with unusual activity
- ☐ Recognized potential attack patterns

- ☐ Successfully drilled down into specific time ranges
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Task 4: Refine Results Using Field Values

Steps:

1. Use Interactive Fields

- Click on a username in the search results
- Select **Add to search** from the popup menu
- Observe updated search string and results

2. Analyze User-Specific Activity

- Review timeline for spikes in this user's failed attempts
- If spike exists, double-click timeline column to zoom in

3. Remove Field Filter

- Click the same username again
- Select **Remove from search**
- Verify search returns to broader results

Validation:

- ☐ Successfully added user to search filter
 - ☐ Search results narrowed to specific user
 - ☐ Timeline shows user-specific activity patterns
 - ☐ Successfully removed filter to return to broader results
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Task 5: Save and Share Search Results

Steps:

1. Access Job Settings

- From **Job** menu (below search bar), select **Edit Job Settings**

2. Modify Permissions

- Change **Read Permissions** from Private to **Everyone**
- This allows others to leverage your search work

3. Extend Lifetime

- Change search lifetime from default (10 minutes) to **7 days**
- Note options to copy link or bookmark search

4. **Save Changes**

- Click **Save** to return to Search view

5. **Review Job History**

- Navigate to **Activity > Jobs** (right side of Splunk bar)
- Review Owner, Events, Expires, Status, and Actions columns
- Click on your search criteria (in blue) to reopen saved search

Validation:

- ☐ Successfully modified job permissions to Everyone
 - ☐ Extended search lifetime to 7 days
 - ☐ Can access job history
 - ☐ Can reopen saved search from history
 - ☐ Search reopens without re-executing
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Progress Tracking Checklist

Basic Search Skills

- ☐ Executed searches with Boolean operators (AND, OR)
- ☐ Used wildcards (*) effectively
- ☐ Applied proper time range selection
- ☐ Navigated paginated results

Search Refinement

- ☐ Filtered results using exact phrases with quotes
- ☐ Added and removed field-based filters
- ☐ Used interactive field selection menus

Analysis Techniques

- ☐ Interpreted timeline patterns
- ☐ Identified potential security threats
- ☐ Correlated events across time periods
- ☐ Recognized attack patterns

Search Management

- ☐ Modified job permissions and lifetime
- ☐ Accessed and managed job history
- ☐ Shared search results appropriately

Key Concepts Summary

Search Language Basics

- **Boolean Operators:** Use AND, OR for combining search terms
- **Wildcards:** Use * for pattern matching (fail* matches fail, failed, failure)
- **Exact Phrases:** Use quotes for exact string matching ("port 22")

Timeline Analysis

- Timeline visualization helps identify patterns and anomalies
- Spikes in activity may indicate security incidents
- Time-based correlation reveals attack sequences

Interactive Search

- Click field values to dynamically modify searches
- Add/remove filters without retyping search strings
- Interactive approach speeds up investigation workflows

Search Job Management

- Control search permissions for collaboration
- Extend job lifetime for important searches
- Job history provides audit trail and reusability

Troubleshooting

Common Issues

Search returns no results:

- Verify time range covers data periods
- Check search syntax for typos
- Ensure data sources are available

Too many results to analyze:

- Add more specific filters
- Use exact phrases with quotes

- Narrow time range to focus investigation

Timeline shows no activity:

- Expand time range
- Verify search terms match data
- Check if events exist in selected timeframe

Cannot save or share searches:

- Verify user permissions
- Check system storage limits
- Confirm job settings access

Windows/Mac Compatibility Notes**Windows Users:**

- Use Command Prompt or PowerShell for any command line operations
- Browser compatibility: Chrome, Firefox, Edge supported
- File paths use backslashes () in Windows

Mac Users:

- Use Terminal for command line operations
- Browser compatibility: Safari, Chrome, Firefox supported
- File paths use forward slashes (/) in macOS

Both Platforms:

- Ensure JavaScript is enabled in browser
- Clear browser cache if interface issues occur
- Check firewall settings allow Splunk web interface access

Cleanup Instructions

1. Optional: Clear Search History

- Navigate to Activity > Jobs
- Delete unnecessary search jobs to free storage
- Keep important searches with extended lifetimes

2. Reset Time Range

- Set time picker back to default range (Last 24 hours)
- This improves performance for future searches

3. Close Browser Tabs

- Close extra Splunk tabs to free resources
- Keep main Search & Reporting tab open if continuing

Validation Steps

After completing this lab, verify your skills by:

1. **Independent Search:** Create a search for successful logins without guidance
2. **Pattern Recognition:** Identify at least one suspicious activity pattern
3. **Search Management:** Save one search with appropriate permissions
4. **Timeline Analysis:** Explain what timeline patterns might indicate security issues

Next Steps

This lab covered fundamental searching techniques. In the next lab, you will learn about:

- Using the Fields sidebar to analyze search results
- Filtering searches using field values and comparison operators
- Navigating and utilizing Search History
- Managing search jobs

Additional Resources

- Splunk Search Language Reference
- Best Practices for Search Performance
- Security Use Cases and Examples
- Advanced Timeline Analysis Techniques